Data Structure Lab9 : Positional Linked List and Array List 2024-2025

Topics

1. Create Position Interface
2. Create Positional List interface
3. Create Positional Linked List Using Linked List structure (Nodes)
4. Implement Basic Methods of Positional Linked List
    - addBefore(Position<E> p ,E e)
    - addAfter(Position<E> p ,E e)
    - remove(Position<E> p)
5. Implement Iterator and Iterable pattern design in Positional Linked Lists

Homework

1. Implement the ArrayList Data structure as it is described in chapter 7.

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```
;import java.util.Iterator
;import java.util.NoSuchElementException

}<public class Lab9<T
;private Object[] array
;private int size

}()public Lab9
;array = new Object[10]
;size = 0
{

}public void add(T element)
}if (size == array.length)
;()resize
{
;array[size] = element
;++size
{

}public T get(int index)
}if (index >= size || index < 0)
;throw new IndexOutOfBoundsException("Index out of bounds")
{
;return (T) array[index]
{

}public void remove(int index)
}if (index >= size || index < 0)
;throw new IndexOutOfBoundsException("Index out of bounds")
{
}for (int i = index; i < size - 1; i++)
;array[i] = array[i + 1]
{
;array[size - 1] = null
;--size
{

}()public int size
;return size
{

}()private void resize
;Object[] newArray = new Object[array.length * 2]
;System.arraycopy(array, 0, newArray, 0, array.length)
;array = newArray
{
```

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2. Implement the iterator idea in your ArrayList.

```
}()public Iterator<T> iterator
;()return new Lab9Iterator
{

}<private class Lab9Iterator implements Iterator<T
;private int currentIndex

}()public Lab9Iterator
;currentIndex = 0
{

Override@
}()public boolean hasNext
;return currentIndex < size
{

Override@
}()public T next
}if (!hasNext())
;throw new NoSuchElementException("No more elements in the list")
{
;return (T) array[currentIndex++]
{
{
{
```